

KRATIKA RATHI

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EDUCATION

Master of Science, Computer Science (Artificial Intelligence)
University of Southern California, Los Angeles, United States

2026 - 2027 (Expected)

Bachelor of Science, Statistics

The University of British Columbia, Vancouver, Canada

2020 - 2025

SKILLS

Algorithm Design and Analysis, Dynamic Programming, Machine Learning and Adaptive Systems, Artificial Intelligence, Bayesian Statistics, Multivariate Statistical Analysis, Data Science, Probability, Statistical Modeling, Time Series and Forecasting

Languages: **Python, R, VBA, SQL, Java, HTML, JavaScript** Software/Development tools: **Tableau, Power BI, MS Access, GitHub, Jira**

EXPERIENCE

Graduate Researcher, Melady Lab at USC

June 2026 - Current

- Designed **STRATA**, a framework for decomposing LLM agent skills into reusable **procedure cores** and environment specific **typed convention adapters**, enabling efficient cross-environment skill transfer.
- Developed a **self-improving** agent framework that revises its written instructions from execution feedback while using evaluation to prevent performance degradation across previously explored environments.
- Curated datasets and developed an evaluation framework to measure how well agent skills transfer across environments.

Actuarial Intern, iA Financial

January 2024 - August 2024

- Worked in the Special Markets Group and performed monthly reserve calculations using the Chain-Ladder model and conducted in-depth analyses to assess their financial viability.
- Created several profitability studies to determine trends and utilized a GARCH model to calculate risk, predict size and frequency of future claims, contributing to a **15% improvement in financial forecasting**.
- Designed, implemented, and validated survey instruments to streamline data collection and quality assurance processes, reducing data entry and validation errors by **10%**.
- Built and analyzed actuarial pricing, reserving, and valuation models using statistical techniques to support risk assessment.

Data Analyst, Orennia

January 2023 - April 2023

Orennia provides an analytics and insights platform for efficient and accurate capital allocation decisions in energy transition spaces.

- Developed classification models guiding client investment decisions across U.S. energy markets, with nearly **93% clients making better capital allocation decisions**.
- Built and maintained Multinomial Logistic Regression models to predict viability of future power plants, leveraging factors like capacity, location, and size to help clients make informed investment decisions.
- Automated forecast generation with custom Python scripts, enhancing modeling efficiency.

Data Analyst, Animal Welfare Program at UBC

May 2022 - December 2022

UBC's Animal Welfare Program is a world-renowned group specializing in research, teaching and outreach programs to improve the lives of animals in human care, including animals living on farms, in shelters, and in laboratories.

- Applied **mixed-effects linear models** to analyze the repeatability and predictability of cattle feeding and drinking behaviors for welfare monitoring.
- Labeled, processed, and analyzed high-volume cattle behavioral data to identify dominance hierarchies between dairy cattle and support early-warning detection of illness, lameness, and missing animals.
- Refactored and optimized Python data-processing pipelines to improve the efficiency of high-volume data analysis while contributing to statistical analysis, visualization, result interpretation, and technical research reporting.

FUTURE PUBLICATIONS

Classifying Mantle Compositions using Geochemical Data from Hawaiian Lavas

- Investigated a machine learning approach for volcanic rock classification by replacing expensive isotope-based analyses with predictive models derived from relatively cheaper and faster trace element geochemistry.
- Implemented an end-to-end **ELT** pipeline, including missing-data imputation, feature engineering, and trained **logistic regression, SVM, random forest and gradient boosting models** to capture discriminative geochemical signatures.
- Such a geochemical analysis helps provide insights into the composition and evolution of Earth's mantle.